BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 122101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 20

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. $\int_1^4 3\sqrt{x}\,dx =$ _____.

(A) 12
(B) 14
(C) 13
(D) 16

Q2. ∫ e^x dx = _____.

(A) e^x + c
(B) a^x + c
(C) x e^x + c
(D) None of these

Q3. If L_1 and L_2 are the two lines having slopes m_1 and m_2 than L_1 and L_2 are perpendicular if _____.

(A) $m_1=m_2$
(B) $m_1.m_2=-1$
(C) $m_1m_2=1$
(D) $m_1\neq m_2$

Q4. Equation of line that exists the y-intercept c=-4 & m=2 is _____.

(A) y=2x-4
(B) y=2x
(C) y=x-4
(D) y=3x-4

Q5. Radius of the circle $x^2+y^2-8x-6y+9=0$ is _____.

(A) 3
(B) 4
(C) 5
(D) 8

Q6. Length of the tangent drawn from the point (2,3) on the circle $x^2+y^2-2x+4y-18=0$ is _____.

(A) $\sqrt{2}$ (B) 3

(C) $\sqrt{3}$ (D) 5

Q7. If the eccentricity of the conic e=1 then the conic is a _____.

(A) Ellipse (B) Hyperbola

(C) Straight line (D) Parabola

Q8. Circle is a special case of an _____.

(A) Tangent line (B) Parabola

(C) Hyperbola (D) Ellipse

Q9. The general solution of differential equation $(dy/dx)^2 = 1-y^2$ is _____.

(A) $y=\sin^{-1} x + c$ (B) $y=\cos x + c$

(C) $y= \pm \sin(x+c)$ (D) $\tan x + c$

Q10. If $f(x,y,z) = x^2y\ e\text{^}(2x) + (x+y-z)^2$ then find f(-1,1,-1) = _____.

(A) $1-e^2$ (B) e^2+1

(C) e^2 (D) e^{-2}

Q11. Let $g(x) = 3x$ then $g^{-1}(x)$ = _____.

(A) x/3 (B) 3/x

(C) $3x^{-1}$ (D) 3x

Q12. If $\log_3 x = 2$ then x = _____.

(A) 2 (B) 9

(C) 3 (D) 2^3

Q13. $\lim(x\to\infty) (1 + 1/x)\text{^}x$ = _____.

(A) a^x (B) e^x

(C) e (D) a^x lna

Q14. Let y = 2/3x-5 is a function then y' = _____.

(A) 6(3x-5) (B) $2/(3x-5)^2$

(C) 3(3x-5) (D) $6/(3x-5)^2$

Q15. If $y = \tan^{-1}(11x)$ then dy/dx = _____.

(A) $11/(1+(11x)^2)$ (B) $1/(1+(11x)^2)$

(C) $11/(1+x^2)$ (D) $11x/(1+x^2)$

Q16. Let y = e^(σx) is a function then dy/dx = _____.

(A) e^(σx)
(B) σe^(σx)
(C) σe^(σx)/x
(D) e^x/σx

Q17. The Equation of tangent at a point p(x_0,y_0) is _____.

(A) $y=mx_0+c$
(B) x/a + y/b = 1
(C) $y-y_0=m(x-x_0)$
(D) $y-y_1=mx$

Q18. Lim(x→+∞) exp(-x) = _____.

(A) 1
(B) +∞
(C) 0
(D) 1

Q19. Let g(x) = |Sin x i - 2 x j + Cos x k| is a scalar function then g'(x) = _____.

(A) Cos x
(B) Cos x i - 2j - Sin x k
(C) Sin x
(D) $4x/(1+4x^2)$

Q20. ∫ $dx/(x^2+16)$ = _____.

(A) $\tan^{-1} x + c$
(B) $\tan^{-1} x/4 + c$
(C) $1/4 \tan^{-1} x/4 + c$
(D) $\tan^{-1} 4x$

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 50

QII. Attempt any Ten Parts out of the following. Each Part carries equal marks.

(i) If f(x) = {x-3 for x<0; 2x+5 for x≥0} Then find f(-1) and f(1).

(ii) Evaluate Limit (√x-1)/(x-1), x→1

(iii) Use first principal rule find f'(x) if f(x) = 3x.

(iv) If $y = (x^2-2)(3x+1)$ then find dy/dx.

(v) Find the Maclaurin series expansion for the function f(x) = 1/(1+x).

(vi) If f(t) is differentiable vector function of t such that F(t) ≠ 0 then show that d/dt [F(t)/|F(t)|] = F'(t)/|F(t)| - [F(t)·F'(t)] |F(t)| / $|F(t)|^3$

(vii) Evaluate ∫ 2^2 √2 dz.

(viii) Evaluate $\int_0^1$ $1/(x^2+1)$ dx.

(ix) Find the perpendicular distance of the point P(3,-1) from the line 5x+12y-16=0.

(x) Find the equation of circle passes through the points (-3,0), (3,4), (6,-2).

(xi) Sketch the Parabola $x^2=2y$.

(xii) Solve the initial value problem $dy/dx = \cos x$, $y(0)=1$

(xiii) Find d^2y/dx^2 for implicit function $x^2+y^2=r^2$.

SECTION "C"

Maximum Marks: 30

Note: Attempt any Three questions of the following. Each question carries equal Marks.

QIII.

(a) Evaluate Limit $\tan^2 x/x$, $x \to 0$

(b) Find the point which is equidistant from the point A(2,1) B(-4,3) and C(-6,5).

QIV.

(a) Evaluate $\int dx/(x^2-1)$.

(b) Evaluate $\int x^2 \ln x \, dx$.

QV.

(a) Use Maclaurin series for e^x to show that 1 + 1/1! + 1/2! + 1/3! + ... is e.

(b) Find the equation of circle which contain the points (2,6), (6,4) and has its center on the line $3x+2y-1=0$.

QVI.

(a) Solve the Homogenous differential equation $dy/dx = (x+y)/y$

(b) If $u = \tan^{-1}((x^2+y^2)/(x+y))$, then show that $x(\partial u/\partial x) + y(\partial u/\partial y) = \sin u \cos u$.